

statsmodels.tsa.statespace.sarimax.SARIMAX

```
class statsmodels.tsa.statespace.sarimax.SARIMAX(
    endog,
    exog=None,
    order=(1, 0, 0),
    seasonal_order=(0, 0, 0, 0),
    trend=None,
    measurement_error=False,
    time_varying_regression=False,
    mle_regression=True,
    simple_differencing=False,
    enforce_stationarity=True,
    enforce_invertibility=True,
    hamilton_representation=False,
    concentrate_scale=False,
    trend_offset=1,
    use_exact_diffuse=False,
    dates=None,
    freq=None,
    missing='none',
    validate_specification=True,
    **kwargs
)
```

[\[source\]](#)

Seasonal AutoRegressive Integrated Moving Average with eXogenous regressors model

Parameters

endog : [array_like](#)

The observed time-series process y

exog : `array_like`, optional

Array of exogenous regressors, shaped `nobs x k`.

order : `iterable` or `iterable` of `iterables`, optional

The (p,d,q) order of the model for the number of AR parameters, differences, and MA parameters. d must be an integer indicating the integration order of the process, while p and q may either be an integers indicating the AR and MA orders (so that all lags up to those orders are included) or else iterables giving specific AR and / or MA lags to include. Default is an AR(1) model: (1,0,0).

seasonal_order : `iterable`, optional

The (P,D,Q,s) order of the seasonal component of the model for the AR parameters, differences, MA parameters, and periodicity. D must be an integer indicating the integration order of the process, while P and Q may either be an integers indicating the AR and MA orders (so that all lags up to those orders are included) or else iterables giving specific AR and / or MA lags to include. s is an integer giving the periodicity (number of periods in season), often it is 4 for quarterly data or 12 for monthly data. Default is no seasonal effect.

trend : `str` {'n','c','t','ct'} or `iterable`, optional

Parameter controlling the deterministic trend polynomial $A(t)$. Can be specified as a string where 'c' indicates a constant (i.e. a degree zero component of the trend polynomial), 't' indicates a linear trend with time, and 'ct' is both. Can also be specified as an iterable defining the non-zero polynomial exponents to include, in increasing order. For example, $[1,1,0,1]$ denotes $a + bt + ct^3$. Default is to not include a trend component.

measurement_error : `bool`, optional

Whether or not to assume the endogenous observations *endog* were measured with error. Default is False.

time_varying_regression : `bool`, optional

Used when an explanatory variables, *exog*, are provided to select whether or not coefficients on the exogenous regressors are allowed to vary over time. Default

is False.

mle_regression : `bool`, optional

Whether or not to use estimate the regression coefficients for the exogenous variables as part of maximum likelihood estimation or through the Kalman filter (i.e. recursive least squares). If *time_varying_regression* is True, this must be set to False. Default is True.

simple_differencing : `bool`, optional

Whether or not to use partially conditional maximum likelihood estimation. If True, differencing is performed prior to estimation, which discards the first $sD + d$ initial rows but results in a smaller state-space formulation. See the Notes section for important details about interpreting results when this option is used. If False, the full SARIMAX model is put in state-space form so that all datapoints can be used in estimation. Default is False.

enforce_stationarity : `bool`, optional

Whether or not to transform the AR parameters to enforce stationarity in the autoregressive component of the model. Default is True.

enforce_invertibility : `bool`, optional

Whether or not to transform the MA parameters to enforce invertibility in the moving average component of the model. Default is True.

hamilton_representation : `bool`, optional

Whether or not to use the Hamilton representation of an ARMA process (if True) or the Harvey representation (if False). Default is False.

concentrate_scale : `bool`, optional

Whether or not to concentrate the scale (variance of the error term) out of the likelihood. This reduces the number of parameters estimated by maximum likelihood by one, but standard errors will then not be available for the scale parameter.

trend_offset : `int`, optional

The offset at which to start time trend values. Default is 1, so that if *trend*='t' the trend is equal to 1, 2, ..., nobs. Typically is only set when the model created by extending a previous dataset.

use_exact_diffuse : `bool`, optional

Whether or not to use exact diffuse initialization for non-stationary states. Default is False (in which case approximate diffuse initialization is used).

****kwargs**

Keyword arguments may be used to provide default values for state space matrices or for Kalman filtering options. See *Representation*, and *KalmanFilter* for more details.

Attributes

measurement_error : `bool`

Whether or not to assume the endogenous observations *endog* were measured with error.

state_error : `bool`

Whether or not the transition equation has an error component.

mle_regression : `bool`

Whether or not the regression coefficients for the exogenous variables were estimated via maximum likelihood estimation.

state_regression : `bool`

Whether or not the regression coefficients for the exogenous variables are included as elements of the state space and estimated via the Kalman filter.

time_varying_regression : `bool`

Whether or not coefficients on the exogenous regressors are allowed to vary

over time.

simple_differencing : `bool`

Whether or not to use partially conditional maximum likelihood estimation.

enforce_stationarity : `bool`

Whether or not to transform the AR parameters to enforce stationarity in the autoregressive component of the model.

enforce_invertibility : `bool`

Whether or not to transform the MA parameters to enforce invertibility in the moving average component of the model.

hamilton_representation : `bool`

Whether or not to use the Hamilton representation of an ARMA process.

trend : `str` {'n','c','t','ct'} or `iterable`

Parameter controlling the deterministic trend polynomial $A(t)$. See the class parameter documentation for more information.

polynomial_ar : `ndarray`

Array containing autoregressive lag polynomial lags, ordered from lowest degree to highest. The polynomial begins with lag 0. Initialized with ones, unless a coefficient is constrained to be zero (in which case it is zero).

polynomial_ma : `ndarray`

Array containing moving average lag polynomial lags, ordered from lowest degree to highest. Initialized with ones, unless a coefficient is constrained to be zero (in which case it is zero).

polynomial_seasonal_ar : `ndarray`

Array containing seasonal moving average lag polynomial lags, ordered from lowest degree to highest. Initialized with ones, unless a coefficient is

constrained to be zero (in which case it is zero).

polynomial_seasonal_ma : `ndarray`

Array containing seasonal moving average lag polynomial lags, ordered from lowest degree to highest. Initialized with ones, unless a coefficient is constrained to be zero (in which case it is zero).

polynomial_trend : `ndarray`

Array containing trend polynomial coefficients, ordered from lowest degree to highest. Initialized with ones, unless a coefficient is constrained to be zero (in which case it is zero).

k_ar : `int`

Highest autoregressive order in the model, zero-indexed.

k_ar_params : `int`

Number of autoregressive parameters to be estimated.

k_diff : `int`

Order of integration.

k_ma : `int`

Highest moving average order in the model, zero-indexed.

k_ma_params : `int`

Number of moving average parameters to be estimated.

seasonal_periods : `int`

Number of periods in a season.

k_seasonal_ar : `int`

Highest seasonal autoregressive order in the model, zero-indexed.

k_seasonal_ar_params : `int`

Number of seasonal autoregressive parameters to be estimated.

k_seasonal_diff : `int`

Order of seasonal integration.

k_seasonal_ma : `int`

Highest seasonal moving average order in the model, zero-indexed.

k_seasonal_ma_params : `int`

Number of seasonal moving average parameters to be estimated.

k_trend : `int`

Order of the trend polynomial plus one (i.e. the constant polynomial would have $k_{trend}=1$).

k_exog : `int`

Number of exogenous regressors.

Notes

The SARIMA model is specified $(p, d, q) \times (P, D, Q)_s$.

$$\phi_p(L)\tilde{\phi}_P(L^s)\Delta^d\Delta_s^D y_t = A(t) + \theta_q(L)\tilde{\theta}_Q(L^s)\zeta_t$$

In terms of a univariate structural model, this can be represented as

$$y_t = u_t + \eta_t$$
$$\phi_p(L)\tilde{\phi}_P(L^s)\Delta^d\Delta_s^D u_t = A(t) + \theta_q(L)\tilde{\theta}_Q(L^s)\zeta_t$$

where η_t is only applicable in the case of measurement error (although it is also used in the case of a pure regression model, i.e. if $p=q=0$).

In terms of this model, regression with SARIMA errors can be represented easily as

$$y_t = \beta_t x_t + u_t$$

$$\phi_p(L) \tilde{\phi}_P(L^s) \Delta^d \Delta_s^D u_t = A(t) + \theta_q(L) \tilde{\theta}_Q(L^s) \zeta_t$$

this model is the one used when exogenous regressors are provided.

Note that the reduced form lag polynomials will be written as:

$$\Phi(L) \equiv \phi_p(L) \tilde{\phi}_P(L^s)$$

$$\Theta(L) \equiv \theta_q(L) \tilde{\theta}_Q(L^s)$$

If *mle_regression* is True, regression coefficients are treated as additional parameters to be estimated via maximum likelihood. Otherwise they are included as part of the state with a diffuse initialization. In this case, however, with approximate diffuse initialization, results can be sensitive to the initial variance.

This class allows two different underlying representations of ARMA models as state space models: that of Hamilton and that of Harvey. Both are equivalent in the sense that they are analytical representations of the ARMA model, but the state vectors of each have different meanings. For this reason, maximum likelihood does not result in identical parameter estimates and even the same set of parameters will result in different loglikelihoods.

The Harvey representation is convenient because it allows integrating differencing into the state vector to allow using all observations for estimation.

In this implementation of differenced models, the Hamilton representation is not able to accommodate differencing in the state vector, so *simple_differencing* (which performs differencing prior to estimation so that the first $d + sD$ observations are lost) must be used.

Many other packages use the Hamilton representation, so that tests against Stata and R require using it along with simple differencing (as Stata does).

If *filter_concentrated = True* is used, then the scale of the model is concentrated out of the likelihood. A benefit of this is that there the dimension of the parameter vector is reduced so that numerical maximization of the log-likelihood function may be faster and more stable. If this option in a model with measurement error, it is important to note that the estimated measurement error parameter will be relative to the scale, and is named “*snr.measurement_error*” instead of “*var.measurement_error*”. To compute the variance of the measurement error in this case one would multiply *snr.measurement_error* parameter by the scale.

If `simple_differencing = True` is used, then the *endog* and *exog* data are differenced prior to putting the model in state-space form. This has the same effect as if the user differenced the data prior to constructing the model, which has implications for using the results:

- Forecasts and predictions will be about the *differenced* data, not about the original data. (while if `simple_differencing = False` is used, then forecasts and predictions will be about the original data).
- If the original data has an `Int64Index`, a new `RangeIndex` will be created for the differenced data that starts from one, and forecasts and predictions will use this new index.

Detailed information about state space models can be found in [1]. Some specific references are:

- Chapter 3.4 describes ARMA and ARIMA models in state space form (using the Harvey representation), and gives references for basic seasonal models and models with a multiplicative form (for example the airline model). It also shows a state space model for a full ARIMA process (this is what is done here if `simple_differencing=False`).
- Chapter 3.6 describes estimating regression effects via the Kalman filter (this is performed if `mle_regression` is `False`), regression with time-varying coefficients, and regression with ARMA errors (recall from above that if regression effects are present, the model estimated by this class is regression with SARIMA errors).
- Chapter 8.4 describes the application of an ARMA model to an example dataset. A replication of this section is available in an example IPython notebook in the documentation.

References

- [1] Durbin, James, and Siem Jan Koopman. 2012. Time Series Analysis by State Space Methods: Second Edition. Oxford University Press.

Methods

`clone`(endog[, exog])

Clone state space model with new data and optionally new specification